

Meerkat-TRIZ Whitepaper v1.1

Meerkat-TRIZ: A TRIZ-Domain LoRA Fine-tune, and the Evaluation Infrastructure That Kept It Honest

🇺🇸 English | 🇨🇳 中文 Meerkat AI · July 2026 · v1.1 (adds the post-release external-judge final review, §5.2) Artifacts: model · benchmark · code

Abstract

We release **Meerkat-TRIZ-v1**, a LoRA adapter (181MB, $r=64$, 0.24% trainable parameters) that fine-tunes Qwen3.6-35B-A3B — a thinking-native hybrid architecture (Gated DeltaNet $\times$ Gated Attention $\times$ 256-expert MoE) — for Chinese TRIZ (Theory of Inventive Problem Solving) question answering. The model is, to our knowledge, the first published LLM fine-tune for the TRIZ domain.

The more consequential contribution is what happened to the evaluation while getting here. Across a five-version, evaluation-driven iteration campaign we discovered and repaired three measurement failures that each reversed or voided conclusions: (1) a **contaminated baseline** — our harness stripped the empty think block required by the thinking-native base, turning 91/100 base responses into unterminated reasoning drafts and inflating the fine-tune’s apparent judge gain from an artifactual **+1.00** to a true **-0.30**; (2) **extreme judge position bias** — a pairwise judge showed a position inconsistency rate of **0.87**, more than twice the ~25pp swing reported in the literature, making dual-order averaging the only valid protocol; and (3) **systematic dual-track divergence** — keyword and judge tracks share only ~10% variance and once disagreed significantly in opposite directions. A fourth caveat was quantified post-release: (4) **same-family judge inflation** — an external-judge final review (Claude / GPT / Gemini, identical protocol, 299 paired items) confirms the improvement’s direction under two of three judges but at roughly one quarter of the same-family magnitude, putting the family-attributable share of the headline number at about three quarters. Under the clean anchor, the honest summary is: on its native v5 gold protocol the release model improves judge-track quality over base — **+0.09 to +0.10, significant under two of three external judges** (the third sees no significant difference); the same-family judge reads **+0.39** [+0.30, +0.49]. The strongest cross-judge consensus is the targeted concept_explanation repair (significant under all three external judges). Keyword coverage is at parity; on the older v4 protocol the model is statistically tied with the strongest internal baseline. We publish the model, the dual gold-set benchmark, and the full evaluation harness with all statistical discipline machine-checked.

1. Introduction

Domain fine-tuning is a chain of decisions, and every link in that chain is forged by an evaluation signal: Is v2 better than v1? Did the targeted data injection work? Should we ship? If the signal is contaminated, every downstream decision is wrong with confidence.

This whitepaper documents a complete iteration campaign — five adapter versions, two benchmark generations, and one public release — in which the evaluation infrastructure itself became the primary object of engineering. We tell the story in both directions: what the measurements said at the time, and what they turned out to mean after each measurement failure was found and repaired. All numbers below come with paired confidence intervals; where the answer was “no significant difference”, we say so.

Release artifacts. Model weights and model card (Meerkat-TRIZ-v1); a dual gold-set benchmark (100 + 300 items); and `meerkat-triz`, the pip-installable evaluation harness whose rules encode every lesson below as a machine-checked assertion rather than a convention.

2. System and Training

Base model. Qwen3.6-35B-A3B (Apache-2.0): 35B total / 3B activated parameters, 40 layers mixing Gated DeltaNet (linear attention) and Gated Attention blocks over a 256-expert MoE, thinking-native chat template.

Adapter. LoRA $r=64$, $\alpha=128$, dropout 0, BF16, targeting twelve projection matrices including the DeltaNet `in_proj_*` family — 84.66M trainable parameters (0.24%). Trained on a single desktop-class unified-memory machine (DGX Spark GB10, 121GB) without quantization. The released v1 model (internally v5a) trained 4 epochs / 5,548 steps on ~11k Chinese TRIZ ChatML SFT samples; final train loss 0.604, eval loss 1.626.

Data pipeline evolution. The five versions are best understood as a data and quality-gate story, not a hyperparameter story: v1 (2.7k samples, no quality gates $\rightarrow$ overall regression), v2 (8.5k multi-angle samples, basic gates), v3 (targeted ARIZ injection), v4 (seven-gate funnel, 11,001 $\rightarrow$ 5,739 samples), v4.1 (single-variable rebalance repairing a `concept_explanation` regression), v5a (dual-style, dual-decomposition, grouped split; empty think block retained — the released model). Hyperparameters were frozen within each line so every comparison had exactly one moving part.

3. Evaluation Methodology

The harness scores every model on two tracks that are never blended:

- **Keyword track:** hit rate of expected terms per item (substring match + human-maintained alias map; pre/post-alias dual scores reported; a miss audit queue watches the alias map itself).
- **Judge track:** an LLM judge (pinned to moonshot-v1-32k, temperature 0 with a hard assertion) scores 0–4 on a four-dimension rubric. Arm A uses an anti-verbosity rubric over untruncated input; Arm B pairs responses against the baseline within same-length buckets to control length bias. A post-release final review re-ran Arm A verbatim with three external judges (§5.2).

Around these tracks sit four disciplines, each enforced by code:

1. **Anchor lineage.** Base-anchor runs must carry a `base_goldfix`-lineage tag or the harness exits (one-vote veto). A dual-anchor consistency check freezes the run if $>10\%$ of items shift $|\Delta kw| > 0.2$ between anchors.
2. **Empty think block retained (E0 rule).** Every prompt is smoke-asserted to retain the `<think>\n\n</think>` block; stripping it is what contaminated our original baseline (§4.1).

3. **Generation quality gates.** Think-residue / non-empty / Chinese-ratio / length floors (plus an English-draft detector); failures get one bad_words_ids fallback regeneration; persistent invalids freeze the evaluation rather than silently entering the scores.
4. **Statistical protocol.** All comparisons are per-item paired: bootstrap (10,000 draws, stdlib Random(42), fingerprint self-check) 95% CIs, Wilson CIs for pass rates, exact McNemar tests. Subsets with $n < 30$ are labeled descriptive and cannot veto. Pairwise judging is AB/BA double-order mandatory; $>10\%$ position inconsistency voids single-order numbers automatically.

4. Three Measurement Failures (and What They Cost Us)

4.1 The contaminated baseline — a +1.00 gain that was actually -0.30

For three versions our reports said the fine-tunes beat base by roughly a full judge point (+1.00 [0.80, 1.19], McNemar $p=1.7e-12$). The signal was false. The harness stripped the empty think block from prompts; the thinking-native base, deprived of its “thinking has ended” anchor, emitted unterminated English reasoning drafts on 91/100 gold items. Both tracks were scoring drafts. After the fix (retain the block, strip closed blocks post-generation), the v4-vs-base judge difference **reversed sign to -0.30 [-0.46, -0.14]**, and the McNemar direction flipped (55/4 $\rightarrow$ 10/30). Every version ranking had to be re-derived; the “keep v2” gate verdict survived, but for entirely different reasons than originally recorded.

4.2 Judge position bias at twice the literature amplitude

A pairwise judge asked to compare responses showed a position inconsistency rate of **0.87** [0.79, 0.92] — the candidate presented second won 81% of the time; on the clean anchor the effect persisted at 0.58 inconsistency. The literature reports ~25pp swings; we measured 62pp. Any single-order pairwise conclusion in this setting is noise. Dual-order merging is now the only valid protocol and is enforced in code. Separately, the judge at T=0 proved **fully deterministic** (0.000 flip rate across repeats — a counterexample to the 13.6% flip rate reported elsewhere), which is what makes our pinned-judge, single-run protocol defensible.

4.3 Dual-track divergence — when the two tracks vote opposite ways

Keyword and judge tracks measure different constructs (~10% shared variance, per-item Spearman $r \approx 0.32$). During iteration they once disagreed significantly in opposite directions: v3 vs v2 scored keyword **+0.0556** (significant) and judge **-0.0682** (significant) — v3 had learned to stuff keywords while semantic quality fell. We also found the keyword track under-detected ARIZ step coverage at a 31.7% miss rate (English/synonymous phrasing), which had fabricated an apparent “v2 ARIZ regression”. Operational rule that survived: **release gates must be dual-track; no single-track verdict ships.**

5. Results

5.1 Release model on the v5 gold protocol (300 items, $n=300$ paired)

Judge-track numbers in this table are **same-family (Moonshot) readings** — see §5.2 for the external-judge final review, which confirms the direction at smaller magnitude.

Metric	base (clean anchor)	Meerkat-TRIZ-v1	Paired diff [95% CI]	
Keyword mean	0.6384	0.6383	-0.0001	[-0.017, +0.017] n.s.
Judge Arm-A mean	3.0300	3.4233	+0.3933	[+0.297, +0.490] sig.
Judge pass rate	0.843	0.947	McNemar p=1.5e-05	

Per-subset judge gains: concept_explanation **+0.733** [+0.467, +1.022], innovation_assessment +0.600, contradiction_analysis +0.467, case_generation +0.378, ariz_guidance +0.317 (all significant); principle_recommendation +0.050 n.s. Quality gates passed 299/300; overrefusal 0/300. Training-set contamination scan against both gold sets: **0 hits** (3-gram Jaccard ≥ 0.5).

5.2 External-judge final review (post-release, 299 paired items)

After release we re-ran the Arm-A judge protocol verbatim — same rubric, same prompts, T=0, same paired bootstrap — with three external judges from three vendors: claude-sonnet-4-6 (Anthropic), gpt-5.4 (OpenAI), gemini-3.5-flash (Google). One item was blocked by the provider content filter under all judges, hence n=299.

Judge	base mean	v1 mean	Paired diff [95% CI]		Sig.
moonshot-v1-32k (same-family reference)	3.030	3.423	+0.393	[+0.297, +0.490]	yes
claude-sonnet-4-6	2.555	2.649	+0.094	[+0.020, +0.167]	yes
gpt-5.4	2.301	2.405	+0.104	[+0.020, +0.184]	yes
gemini-3.5-flash	2.871	2.823	-0.048	[-0.144, +0.045]	no

How to read this:

- **Direction confirmed, magnitude not.** Two of three external judges preserve the positive direction with statistical significance, at roughly one quarter of the same-family magnitude; the third sees no difference. The defensible headline is “+0.09 ~ +0.10, significant under two of three external judges”; roughly three quarters of the same-family +0.39 is attributable to judge-family effects and must not be extrapolated.
- **It is a scoring-system difference, not judge noise.** The external judges agree with each other at per-item Spearman **0.63–0.75**, but each agrees with moonshot at only **0.27–0.31**.

- **concept_explanation is the strongest consensus** — the subset the v4.1 rebalance targeted gains **+0.24 / +0.31 / +0.29, significant under all three external judges**. The repair is real and externally verified.
- **Watch item.** Gemini scores principle_recommendation significantly lower for the release model (−0.31 [−0.53, −0.10]); the other two judges see no significant difference there. With single-judge support we log this as an open signal for the next data iteration, not an established regression.

Full report and raw data: EXTERNAL_JUDGE_REVIEW.md (中文) · external_review_result.json.

5.3 Six-way comparison on the v4 gold protocol (100 items)

Model	Keyword	Judge	Judge pass
base (legacy, think-contaminated — reference only)	0.3661	1.5700	0.120
v2 (clean anchor)	0.5483	2.5800	0.620
v3	0.5716	2.2800	0.430
v4	0.5568	2.5700	0.630
v4.1	0.5289	2.6200	0.630
Meerkat-TRIZ-v1 (v5a)	0.5391	2.3600	0.600

v5a vs v2 paired: keyword −0.0093 [−0.062, +0.039] n.s.; judge −0.2200 [−0.470, +0.010] n.s. — **a statistical tie with the strongest internal baseline**. The v4 harness strips the empty think block (a format mismatch against v5a’s training format), biasing the judge direction negative; the tie is therefore a lower-bound reading. The v4.1 rebalance repaired the concept_explanation regression without significant cost (kw diff vs v2 −0.0397 [−0.103, +0.025], CI includes 0).

5.4 What fine-tuning actually did

Under the clean anchor the true effect of the earlier fine-tunes was behavioral compression: think-style verbose answers (~3,250 chars) became direct answers ~1/10 the length at parity keyword coverage, with judge-track quality slightly below the verbose base. The v5a line is the first to convert that compression into a genuine judge-track gain on its native protocol — while remaining, on the older protocol, only tied with v2. Both facts are in the model card. We did not run external-model comparisons and claim none; the final review in §5.2 varies the **judge**, not the compared models.

6. Limitations

- **Judge family.** Data generator and primary judge share the Moonshot family (declared “weak same-origin”). The post-release external review (§5.2) confirms the improvement’s direction under two of three external judges but at ~1/4 the magnitude — same-family absolute scores should not be extrapolated across judge families. One external judge (Gemini) detects no overall

gain and a significant principle_recommendation decrease; with single-judge support we treat it as an open signal, not an established regression.

- **Power.** 100-item subsets ($n=10-20$) are descriptive only; “tie with v2” means “no significant difference detected”, not proven equivalence.
- **Length confound.** Base answers are $\sim 10\times$ longer; the judge’s completeness dimension naturally favors length. Arm B (same-length buckets) had too few pairs (17/300) to fully control this.
- **Single domain, single base, single PEFT config.** TRIZ Chinese QA only; no target-module ablation.
- **Benchmark release shape.** Reference answers are withheld from the public benchmark (copyright); the keyword track is fully reproducible, judge-track absolute scores are not.

7. Reproducibility and Artifacts

- Model: <https://huggingface.co/Meerkat-AI/Meerkat-TRIZ-v1> (adapter + model card with full claim discipline)
- Benchmark: <https://huggingface.co/datasets/Meerkat-AI/triz-gold-benchmark> (question + keywords; CC-BY-NC-4.0)
- Code: <https://github.com/coidea-sys/meerkat-triz> — harness, frozen training config (configs/train_v5a.json), methodology docs, CI that runs the unit tests, a no-GPU dry-run of the full pipeline, and the decision-gate backtest on every push.

Determinism: greedy generation, T=0 judge (measured flip rate 0.000), stdlib Random(42) bootstrap, config snapshots embedded in every result.

8. Citation

```
@misc{meerkat-triz-whitepaper-2026,  
  title = {Meerkat-TRIZ: A TRIZ-Domain LoRA Fine-tune, and the Evaluation  
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  year  = {2026},  
  url   = {https://github.com/coidea-sys/meerkat-triz}  
}
```

A venue-formatted academic version of the measurement-failure case study (working title: “When the Baseline Lies”) is in preparation; this whitepaper is the release-supporting document and the single source for citable numbers.